Student Habits and Academic Performance

Summary of Analysis, Key Findings, and Practical Solutions

This document summarizes the analysis of the student_habits_performance dataset. The aim was to understand how students' daily habits relate to examination performance, identify areas that may need attention, and recommend realistic actions for students and schools.

1. Overview of the Dataset

The dataset contains 1,000 student records and 16 original columns. It includes study hours, social media use, Netflix use, sleep duration, attendance, diet quality, exercise frequency, mental health ratings, internet quality, part-time employment, and examination scores.

Average examination score	**69.60 out of 100**
Average daily study time	3.55 hours
Average daily social media use	2.51 hours
Average daily Netflix use	1.82 hours
Average sleep duration	6.47 hours
Average attendance	84.13%
Missing parental education records	91 (9.1% of students)
Duplicate records	0 found

2. Main Findings

Study habits and examination performance

Study hours showed the strongest observed relationship with exam scores (correlation of approximately +0.83). Students who reported studying longer generally achieved higher scores. This is an association and does not prove that study time alone causes higher performance.

Social media and Netflix use

Both measures showed weak negative correlations with exam scores (approximately -0.17 each). This suggests heavier entertainment screen use was associated with slightly lower scores in this dataset, but it does not establish cause and effect.

Sleep duration

Students slept an average of about 6.47 hours. The relationship between sleep and exam scores was weakly positive (approximately +0.12). This supports reviewing sleep routines while recognizing that individual sleep needs vary.

Mental wellbeing

Mental health ratings had a moderate positive relationship with exam scores (approximately +0.32). Students with higher recorded ratings tended to score higher. These ratings should not be treated as a clinical diagnosis.

Attendance

Average attendance was about 84.13%, while its correlation with exam scores was weak (approximately +0.09). Attendance is worth monitoring because it affects opportunities to learn, but it does not explain performance on its own.

Differences in academic performance

Using report-specific categories, 131 students scored below 50, 358 scored from 50 to below 70, 323 scored from 70 to below 85, and 188 scored 85 or above. These are analytical groupings rather than official grades.

Other factors and data limitations

The dataset recorded 162 students with poor internet quality and 215 students with part-time jobs. These circumstances may be relevant, but the analysis does not establish that they caused lower scores. There were 91 missing parental-education entries, limiting analysis of that variable.

3. Problems Identified and Solutions Provided

Problem or area for attention	Practical solution
Uneven study routines	Help students create realistic timetables, use active recall and spaced revision, practise past questions, and seek help with difficult subjects.
Potentially distracting screen time	Encourage students to track screen use, silence non-essential notifications during study, and plan recreational activities around academic priorities.
Short average sleep duration	Promote consistent bedtimes and waking times, reduce late-night screen use, and avoid relying on all-night revision. Most adults need 7–9 hours; teenagers may need more.
Students struggling academically	Offer diagnostic assessments, subject-specific remedial lessons, tutoring, regular feedback, and individual progress targets.
Wellbeing concerns	Make counselling and trusted support available, teach manageable planning and stress-management skills, and maintain a supportive, non-stigmatizing environment.
Attendance difficulties	Monitor patterns of absence, investigate barriers before responding, communicate with students and families where appropriate, and provide catch-up materials.
Unequal access to internet or learning resources	Provide downloadable or printed materials, school library/computer access, and offline alternatives for essential learning tasks.
Missing data	Investigate why parental education information is absent; recover it only when appropriate and available, otherwise preserve missing values and report the limitation.
Different student needs	Use differentiated support: remedial help for struggling students, structured practice for students who need

	improvement, and extension opportunities for high performers.

4. Suggested School Action Plan

Period	Suggested action
Week 1	Record baseline scores, attendance, study routines, and support needs.
Week 2	Help students build realistic personal study timetables.
Weeks 3–4	Begin targeted revision, tutoring, and healthy sleep/screen-time habit tracking.
Weeks 5–6	Check progress and provide extra support where difficulties remain.
Week 7	Conduct a practice assessment and review common learning gaps.
Week 8	Compare results with the baseline and adjust the support plan.

Progress can be monitored through changes in examination scores, the number of students scoring below 50, attendance, study routines, sleep habits, participation in support sessions, and student feedback. Where feasible, comparisons with similar students who did not receive an intervention can help assess whether the programme contributed to improvement.

5. Conclusion

The analysis suggests that study time has the strongest observed relationship with examination scores in this dataset. Mental wellbeing also shows a meaningful positive association, while sleep, attendance, social media use, and Netflix use show weaker relationships. The findings point toward a balanced approach: effective study strategies, sensible time management, sufficient rest, wellbeing support, accessible learning resources, and targeted academic assistance.

The results should guide further investigation, not be treated as proof that any single habit causes academic success or failure. The most useful next step is to apply practical interventions, monitor progress, and adjust support based on students' actual needs.